



Newington College

2009

TRIAL HSC EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is proved at the back of this paper
- All necessary working should be shown in every question

Total marks: 120

- Attempt Questions 1–10
- All questions are of equal value

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Total Marks - 120

Attempt Questions 1 – 10.

All questions are equal value.

Answer each question in a SEPARATE writing booklet.

Question 1 (12 Marks) Use a SEPARATE writing booklet.

Marks

(a) Find the value of $\sqrt{\frac{50}{\pi}}$ correct to three significant figures. 2

(b) Factorise $3x^2 + 5x - 12$. 2

(c) Simplify $\frac{3}{5} - \frac{x-1}{4}$. 2

(d) Find the primitive of $4 - \frac{1}{x^2}$. 2

(e) Find the values of x for which $|x-4| > 3$. 2

(f) If $f(x) = \begin{cases} 7-3x & \text{for } x \leq 1 \\ x^3+1 & \text{for } x > 1 \end{cases}$
evaluate $f(0) + f(2)$. 2

Question 2 (12 Marks) Use a SEPARATE writing booklet.

(a) Differentiate the following:

(i) $(2x^3 - 1)^5$ 2

(ii) $e^{5x} \tan x$ 2

(iii) $\frac{\ln x}{x^2}$ 2

(b) Find the equation of the tangent to the curve $y = 2x(1-x^2)$ at the point $(1, 0)$.

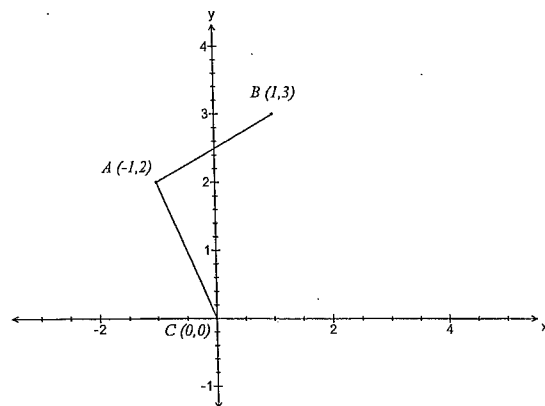
3

(c) Evaluate $\int_2^3 \frac{3x}{x^2-1} dx$, give answer in exact form. 3

Question 3 (12 Marks) Use a SEPARATE writing booklet

Marks

(a)

In the diagram above $A(-1,2)$, $B(2,3)$ and $C(0,0)$ are as shown.

- (i) Prove that $AB \perp AC$. 2
- (ii) Find the equation of CD such that $CD \parallel AB$. 2
- (iii) If BD parallel to the x-axis find the co-ordinates of D . 2
- (iv) Describe the special quadrilateral $ABDC$ geometrically. 1
- (v) Find the area of $ABDC$. 3

(b) The table shows values of a function $f(t)$ for the five values of t .

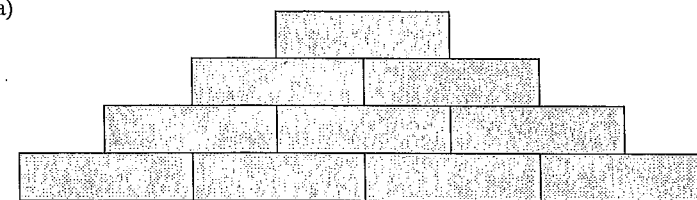
t	0	3	6	9	12
$f(t)$	0	2	5	8	4

Use the Trapezoidal Rule with these 5 values to estimate $\int_0^{12} f(t) dx$. 3

Question 4 (12 Marks) Use a SEPARATE writing booklet.

Marks

(a)



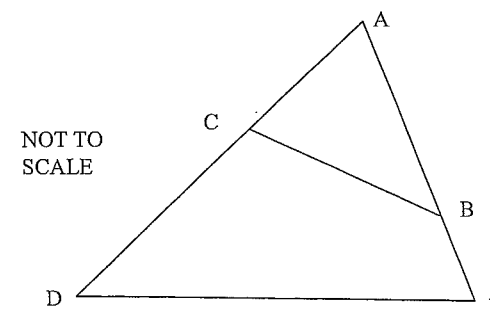
Top four layers.

A bricklayer builds a triangular wall. The top of the wall is shown above. If the bricklayer uses 171 bricks, how many layers did he build?

3

(b) Sally has a bag of sweets which are all identical in shape. The bag contains 6 orange drops and 4 lemon drops. She selects one sweet at random, eats it, and then takes another at random. Determine the probability that:

- (i) both sweets were lemon drops 2
- (ii) at least one of the drops was an orange drop. 2

(c) Evaluate $\sum_{n=1}^{10} 2^n$. 3(d) In the diagram below, $\triangle ABC$ is similar to $\triangle ADE$. $\angle ADE = \angle ABC$, $AC = 8\text{ cm}$, $AB = 12\text{ cm}$ and $BE = 4\text{ cm}$ Find the length of CD . 2

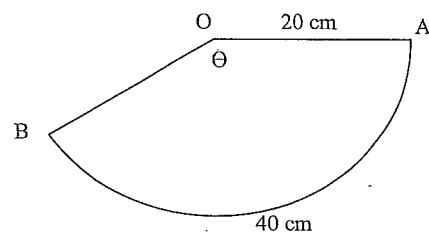
Question 5 (12 Marks) Use a SEPARATE writing booklet.

Marks

(a) Solve $\log_3(5x+1) = 2$.

2

(b)



In the diagram AB is an arc of a circle with centre O and radius 20 cm. The arc has a length of 40 cm. Find:

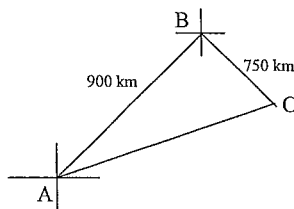
(i) the exact size of $\angle AOB$ in radians,

1

(ii) the exact area of the sector AOB .

1

(c) A plane travels 900 km from A to B on a bearing of $50^\circ T$. The plane then travels 750 km on a bearing of $135^\circ T$ to a point C .



Copy the diagram into your answer booklet.

(i) Show that $\angle ABC$ is 95° , giving reasons.

2

(ii) Find the distance of CA , correct to one decimal place.

2

(iii) Find $\angle ACB$.

2

(iv) Find the bearing of A from C , to the nearest minute.

2

Question 6 (12 Marks) Use a SEPARATE writing booklet.

Marks

(a)

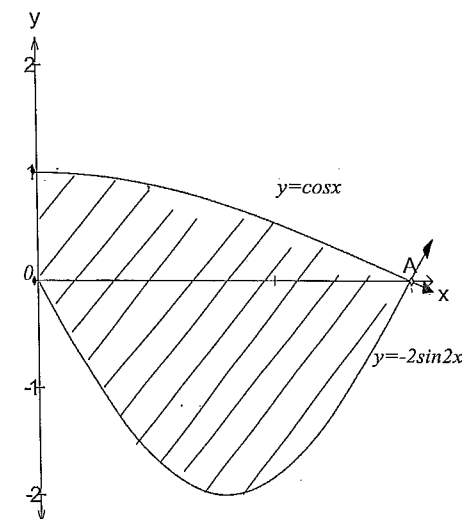


DIAGRAM NOT DRAWN TO SCALE

The diagram shows the graph of $y = -2 \sin 2x$ and $y = \cos x$, for $x \geq 0$. The graphs intersect at point A .

(i) Show that point A has co-ordinates $\left(\frac{\pi}{2}, 0\right)$.

2

(ii) Find the shaded area enclosed by the two graphs for $0 \leq x \leq \frac{\pi}{2}$.

3

(b) Given the equation $2x^2 + 7x - 3 = 0$ has roots α and β , evaluate the following:

(i) $\alpha + \beta$

1

(ii) $\alpha \beta$

1

(iii) $2\alpha^2 + 2\beta^2$.

2

(c) Find the values of A , B and C , such that:

$$4x^2 + 5x - 3 \equiv A(x-1)^2 + B(x-1) + C$$

3

Question 7 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Given the parabolic equation:

$$8y = x^2 - 2x - 15$$

Find:

- (i) the coordinates of the vertex

2

- (ii) the coordinates of the focus

1

- (iii) the equation of the directrix.

1

- (b) For the curve $f(x) = 2x^3 + 3x^2 - 12x - 4$

- (i) Find the coordinates of the stationary points and determine their nature.

3

- (ii) Find the coordinates of the point of inflexion.

2

- (iii) Hence sketch the graph of $y = f(x)$, showing the turning points, the point of inflexion and where the curve meets the y -axis.

2

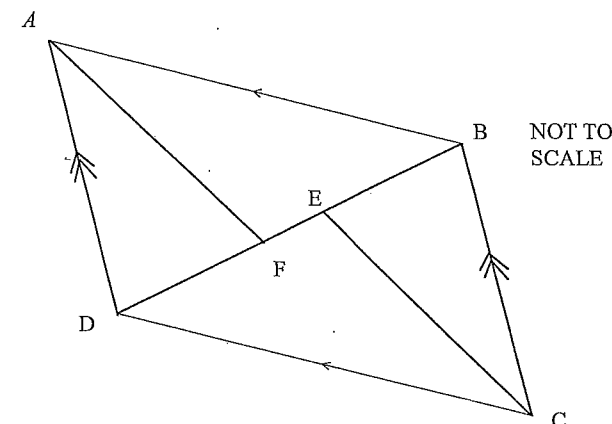
- (iv) For what values of x is the graph of $f(x)$ concave up?

1

Question 8 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Copy or trace the diagram into your writing booklet.



$ABCD$ is a parallelogram. FA bisects $\angle BAD$ and EC bisects $\angle BCD$.

- (i) Prove that $\triangle ADF$ and $\triangle CBE$.

3

- (ii) Hence find the length of EF if $BD = 40\text{ cm}$ and $DF = 16\text{ cm}$.

1

- (b) The position $x\text{ cm}$ of a particle P moving along the x -axis after t seconds is given by $x = 25t - 10\ln t\text{ cm}$, where $t \geq 1$.

- (i) Find expressions for the particle's velocity and acceleration.

2

- (ii) Find the velocity and acceleration when $t = e$ minutes.

2

- (iii) Discuss the velocity as $t \rightarrow \infty$.

2

- (iv) Sketch the graph of the velocity function?

2

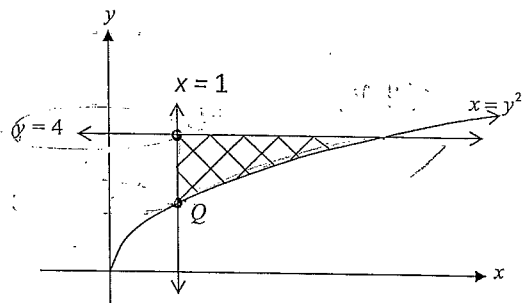
Question 9 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Solve $4\cos^2\theta - 3 = 0$, if $0 \leq \theta \leq 2\pi$.

3

(b)



- (i) Find the coordinates of Q , the intersection of $x = y^2$ and $x = 1$.
- (ii) The shaded region in the diagram is the area bounded by the lines $y = 4$, $x = 1$ and the parabola $x = y^2$. This region is rotated about the y -axis. Find the volume of the solid formed.

1

3

- (c) The amount Q grams of a carbon isotope in a dead tree trunk is given by $Q = Q_0 e^{-kt}$ where Q_0 and k are positive constants and time t is measured in years from the death of the tree.

- (i) Show that Q satisfies the equation $\frac{dQ}{dt} = -kQ$

1

- (ii) Show that if the half-life of the isotope is 4500 years,
then $k = \frac{1}{4500} \ln 2$

2

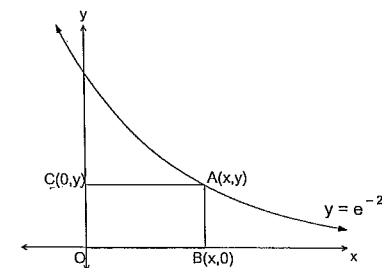
- (iii) For a particular dead tree trunk, the amount of isotope is only 15% of the original amount in the living tree. How long ago did the tree die? Give your answer to the nearest 1000 years.

2

Question 10 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) A rectangle is placed under the curve $y = e^{-2x}$ as shown. If $A(x)$ gives the area of $OBAC$,



- (i) find an expression for $A(x)$ in terms of x .
- (ii) determine the coordinates of $A(x)$ such that the rectangle has a maximum area.

2

3

- (b) A couple wish to purchase a home. They need to calculate the maximum amount they can borrow given that the current interest rate is 9% pa, compounded monthly and the term of the loan is 30 years. They can afford to repay \$2700 per month, paid at the end of every month. Let A_n be the amount owing at the end of n months and P be the amount borrowed.

- (i) Show that after three months the amount of money owing is

$$A_3 = P(1.0075)^3 - 2700(1 + 1.0075 + 1.0075^2).$$

2

- (ii) Hence show that if the loan is to be paid off after n months then

$$P = \frac{360000(1.0075^n - 1)}{(1.0075)^n}$$

2

- (iii) Calculate, correct to the nearest cent, the amount that they can borrow if the loan is to be paid off in thirty years.

2

- (iv) How much interest, in dollars, will they pay for loan?

1

End of Paper.

21 (a) $\sqrt{\frac{50}{17}} = 3.989422806 \dots$ ✓
 $= \underline{3.99}$ [3 sig fig] ✓ 2 for correct answer only

(b) $3x^2 + 5x - 12$ ✓
 $= (3x - 4)(x + 3)$ ✓

(c) $\frac{3}{5} - \frac{x-1}{6}$ ✓
 $= \frac{12 - 5(x-1)}{20}$ ✓
 $= \frac{12 - 5x + 5}{20}$ ✓
 $= \frac{17 - 5x}{20}$ ✓

(d) $\int 4 - \frac{1}{x^2} dx = \int 4 - x^{-2} dx$ ✓
 $= 4x + x^{-1} + C$ ✓
 $= 4x + \frac{1}{x} + C$ ✓

(e) $|x-4| > 3$
 $\therefore x-4 > 3$ or $x-4 < -3$
 $\therefore x > 7$ or $x < 1$ ✓ ✓

(f) $f(0) = 7 - 3(0) = 7$
 $f(2) = 2^3 + 1 = 9$
 $\therefore f(0) + f(2) = 7 + 9 = 16$ ✓ ✓

Q2 (a) (i) $y = (2x^3 - 1)^5$ 12

$\frac{dy}{dx} = 5(2x^3 - 1)^4 \times 6x^2$
 $= 30x^2(2x^3 - 1)^4$ ✓ ✓

(ii) $y = e^{5x} \tan x$
 $\frac{dy}{dx} = e^{5x} \sec^2 x + \tan x \cdot 5e^{5x}$ ✓ ✓ Product Rule.
 $= e^{5x} (\sec^2 x + 5 \tan x)$

(iii) $y = \frac{\ln x}{x^2}$ Quotient Rule
 $\frac{dy}{dx} = \frac{x^2 \times \frac{1}{x} - \ln x \times 2x}{x^4}$ ✓
 $= \frac{x - 2x \ln x}{x^4}$
 $= \frac{1 - 2 \ln x}{x^3}$ ✓

(b) $y = 2x(1-x^2)$
 $\therefore y = 2x - 2x^3$
 $\frac{dy}{dx} = 2 - 6x^2$
 at $x=1$ $\frac{dy}{dx} = 2 - 6 = -4$ ✓
 $\therefore$ Equation of tangent at (1, 0)
 $y - y_1 = m(x - x_1)$ ✓
 $y - 0 = -4(x - 1)$
 $y = -4x + 4$ ✓
 $\therefore 4x + y - 4 = 0$ is the equation of tangent. Note: Accepted form

Q2(c)

$$\int_2^3 \frac{3x}{x^2-1} dx$$

$$= \frac{3}{2} \int_2^3 \frac{2x}{x^2-1} dx \quad \checkmark$$

$$= \frac{3}{2} [\ln(x^2-1)]_2^3 \quad \checkmark$$

$$= \frac{3}{2} [\ln 8 - \ln 3] \quad \checkmark \quad \text{Accept either.}$$

$$= \frac{3}{2} \ln \frac{8}{3} \quad \checkmark$$

Question 3

$$\begin{aligned} \text{a) } M_{AB} &= \frac{3-2}{1-1} & M_{AC} &= \frac{2}{-1} \\ &= \frac{1}{2} & &= -2 \end{aligned}$$

$$\text{Since } M_{AB} \times M_{AC} = -1 \quad (2)$$

$AB \perp AC$

$$\begin{aligned} \text{ii) } y &= mx \text{ as } b=0 \\ \therefore \text{Equation of CO is } y &= \frac{1}{2}x \quad (1) \end{aligned}$$

$$\begin{aligned} \text{iii) at } O \quad y &= 3 \\ \therefore 3 &= \frac{1}{2}x \\ x &= 6 \quad (2) \\ \therefore O &= (6, 3) \end{aligned}$$

$$\text{iv) ABCD is a Trapezium} \\ \text{(Right } \angle) \quad (1)$$

$$\begin{aligned} \text{v) } A &= \frac{1}{2} \times 6 \times 3 - \frac{1}{2} \times \frac{1}{2} \times 1 + \frac{1}{2} \times 2.5 \times 1 \\ &= 10 \text{ units}^2 \end{aligned}$$

(2)

$$\text{b) } \int_a^b f(x) dx = \frac{1}{2} [f(a) + f(b)]$$

$$\begin{aligned} &= \frac{3}{2} [f(0) + f(3) + f(3) + f(6) + f(6) + f(9) + f(9) + f(12)] \\ &= \frac{3}{2} [0 + 2 + 2 + 5 + 5 + 8 + 8 + 9] \\ &= 51 \quad (3) \end{aligned}$$

Question 4

a) $S_n = 171$
 $d = 1$ $a = 1$
 $\frac{n}{2} [2a + (n-1)d] = 171$
 $\frac{n}{2} [2 + (n-1)] = 171$
 $n(n+1) = 342$
 $n^2 + n = 342$
 $n^2 + n - 342 = 0$
 $(n+19)(n-18) = 0$
 $n = -19$ $n = 18$
 $\therefore n = 18$ as $n > 0$
 $\therefore 18$ layers built (3)

b) $P(U) = \frac{4}{10} \times \frac{3}{9}$
 $= \frac{2}{15}$ (2)

ii) $P(\text{At least 1 Orange}) = 1 - P(U)$
 $= 1 - \frac{2}{15}$
 $= \frac{13}{15}$ (3)

c) $\sum_{n=1}^{\infty} 2^n = 2^1 + 2^2 + 2^3 + 2^4 + \dots + 2^{10}$
 $= 2046$ (3)

OR $= S_n \text{ GP}$
 $= \frac{a(1-r^n)}{1-r}$
 $= \frac{2(1-2^{10})}{1-2}$
 $= 2046$

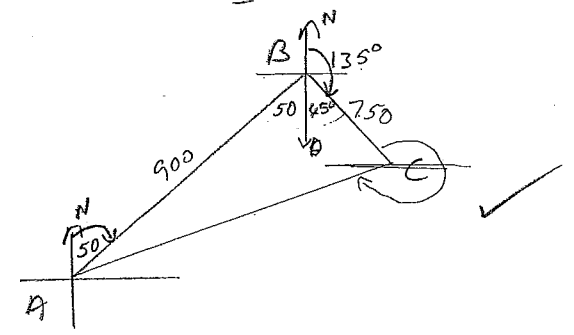
d) $\frac{AD}{12} = \frac{16}{8}$
 $AD = 12 \times \frac{16}{8}$
 $AD = 24$ (2)
 $\therefore CD = 24 - 8$
 $= 16 \text{ cm}$
 OR
 $\frac{x+8}{12} = \frac{16}{8}$
 $x+8 = 12 \times \frac{16}{8}$
 $x+8 = 24$
 $x = 16$
 $\therefore DC = 16 \text{ cm}$

Q 5(a) $\log_3(5x+1) = 2$
 $5x+1 = 9$ ✓
 $5x = 8$
 $x = \frac{8}{5}$ ✓

(b) i) Arc length = $r\theta$
 $\therefore 20\theta = 40$
 $\theta = 2 \text{ radians}$ ✓

(ii) Area of Sector = $\frac{1}{2} r^2 \theta$
 $= \frac{1}{2} (20)^2 \times 2$
 $= 400 \text{ cm}^2$ ✓

(c)

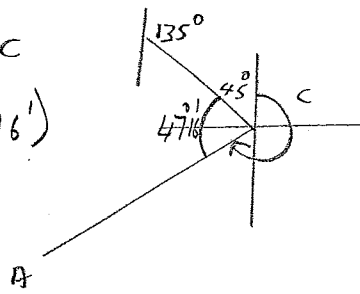


(i) From diagram $\angle ABD = 50^\circ$ [Alternate angles]
 $\angle DBC = 45^\circ$ [Supplementary angles] ✓
 $\therefore \angle ABC = 95^\circ$ Note: Working can be shown on diagram.

(ii) $CA^2 = BC^2 + AB^2 - 2 \times 750 \times 900 \times \cos 95^\circ$ ✓
 $CA = 1220.7 \text{ km}$ [correct to one decimal place]
 ✓ [Note: DO NOT DECREASE FOR ROUNDING]

(iii) $\frac{\sin \angle ACB}{900} = \frac{\sin 95}{1220.7}$
 $\sin \angle ACB = \frac{900 \times \sin 95}{1220.7}$
 $\angle ACB = 47^\circ 16'$ [Note: Accept 47°]

50) (iv) Bearing of A from C
 $= 360^\circ - (45^\circ + 47'16'')$
 $= 267^\circ 44'$



(a) (i) If $y = -2\sin 2x$, $x = \frac{\pi}{2}$
 $\therefore y = -2\sin \pi$
 $= 0$
 $\therefore$ curves intersect at $A(\frac{\pi}{2}, 0)$

If $y = \cos x$, $x = \frac{\pi}{2}$
 $y = \cos \frac{\pi}{2}$
 $= 0$

(ii) Enclosed Area = $\int_0^{\frac{\pi}{2}} [\cos x - (-2\sin 2x)] dx$ ✓
 $= \int_0^{\frac{\pi}{2}} [\cos x + 2\sin 2x] dx$
 $= [\sin x - \cos 2x]_0^{\frac{\pi}{2}}$ ✓
 $= 1 - (-1) - (0 - 1)$
 $= 3 \text{ u}^2$ ✓

(b) $2x^2 + 7x - 3 = 0$

(i) $\alpha + \beta = -\frac{7}{2}$ ✓

(ii) $\alpha\beta = -\frac{3}{2}$ ✓

(iii) $2\alpha^2 + 2\beta^2$
 $= 2(\alpha^2 + \beta^2)$
 $= 2[(\alpha + \beta)^2 - 2\alpha\beta]$ ✓
 $= 2\left[\frac{49}{4} + 3\right]$
 $= 30.5$ ✓

Q6(a)
 (i) $y = -2 \sin x$
 value $x = \frac{\pi}{2}$

Q6(c) $4x^2 + 5x - 3 \equiv A(x-1)^2 + B(x-1) + C$
 Now $A = 4$ [coefficient of x^2]

Let $x = 1 \therefore 6 = C$

Let $x = 0$

$\therefore -3 = A - B + C$

$-3 = 4 - B + 6$

$B = 13$

$\therefore A = 4, B = 13, C = 6$

$4x^2 + 5x - 3 \equiv A(x-1)^2 + B(x-1) + C$
 $= A(x^2 - 2x + 1) + Bx - B + C$
 $= Ax^2 + (B - 2A)x + (A - B + C)$

$A = 4$

$B - 2A = 5$

$B - 2(4) = 5$

$B = 13$

$A - B + C = -3$

$4 - 13 + C = -3$

$-9 + C = -3$

$C = 6$

Q6 (c) $4x^2 + 5x - 3 = A(x-1)^2 + B(x-1) + C$
 let $x=1 \therefore 6 = C$
 let $x=0$

Q7

a) i) $8y = x^2 - 2x - 15$

(ii) $4a = 8$
 $a = 2$

focus $(1, 0)$ ✓

$x^2 - 2x + 1 = 8y + 16$ ✓

$(x-1)^2 = 8(y+2)$ ✓

Vertex $(1, -2)$ ✓

iii) ~~$y = m(x-1)$~~
 $y = m(x-1)$

b) i) $f'(x) = 6x^2 + 6x - 12$

$f''(x) = 12x + 6$

Stationary pts $f'(x) = 0$
 $6x^2 + 6x - 12 = 0$

$x^2 + x - 2 = 0$

$(x+2)(x-1) = 0$

$x = -2 \text{ or } 1$ ✓

$f''(-2) = -24 + 6 < 0$

$\therefore$ max at $(-2, 16)$ ✓

$f''(1) = 12 + 6 > 0$

$\therefore$ min at $(1, -11)$ ✓

ii)

$f''(x) = 0$

$12x + 6 = 0$

$x = (-\frac{1}{2}, 2\frac{1}{2})$

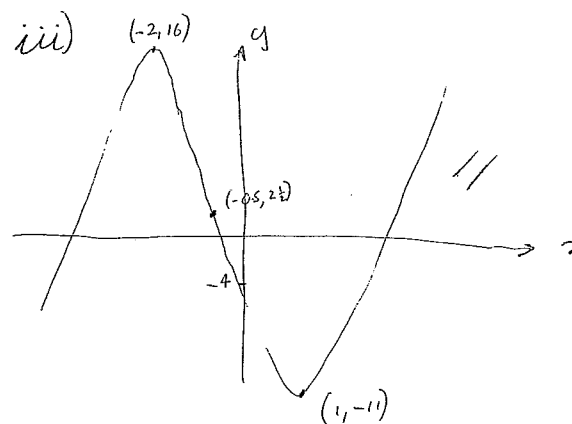
change of concavity
 $\therefore$ point of inflection
 at $(-0.5, 2\frac{1}{2})$

check x

-2	-1/2	1	
y''	< 0	0	> 0

 ✓

iii)



iv) $x > -0.5$

Q8

Q1(a)(i) Consider $\triangle ADF$ and $\triangle CBE$. $AD = CB$ (opposite sides of parallelogram equal) ✓ $\angle ADF = \angle CBE$ (alternate angles $AD \parallel BC$)Now $\angle DAB = \angle BCE$ (opposite angles of parallelogram equal)and since AF bisects $\angle DAB$ and EC bisects $\angle BCE$. $\angle DAF = \angle BCE$ $\therefore \triangle ADF \equiv \triangle CBE$ (AAS) ✓(ii) $DF = EB = 16$ (corresponding sides in congruent triangles)

$$\therefore EF = 40 - 32 = 8 \text{ cm} \quad \checkmark$$

$$(b) (i) v = \frac{dx}{dt} = 25 - \frac{10}{t} \quad \checkmark$$

$$(ii) a = \frac{dv}{dt} = \frac{10}{t^2} \quad \checkmark$$

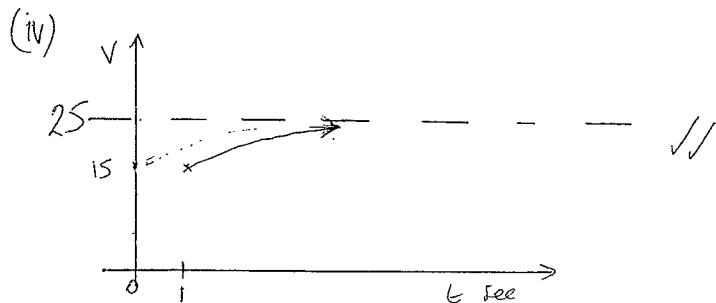
(ii) since t in seconds find $t = 60e$ seconds.

$$\therefore v = 25 - \frac{10}{60e}$$

$$= 25 - \frac{1}{6e} \quad \checkmark \quad (24.94)$$

$$a = \frac{10}{(60e)^2}$$

$$= \frac{1}{360e^2} \quad \checkmark \quad (0.00038)$$

(iii) As $t \rightarrow \infty$ $v \rightarrow 25$ (as $\frac{10}{t} \rightarrow 0$) ✓✓

Q8

Problems with Q1.

(a)(i) On the whole this was done quite poorly.

- some proved similarity using equiangular.
- many tried to show $AF = EC$ by converse reasoning
- language used was incomplete (eg alternate angles on parallel lines).
- many used given for reasons when inappropriate.

(ii) the majority got this correct, regardless of (i).

(b)(i) Most got these, but a few differentiation issues (eg with negative sign & second term)

(ii) few candidates realised they had to substitute $t = 60e$ secs.(iii) Many just wrote $v \rightarrow 25$ with no explanation - two marks usually requires more. Here they needed to show $\frac{10}{t} \rightarrow 0$.

(iv) - Many did not attempt.

- Those that did made a number of errors such as

- starting graph at $t = 0$
- not putting in asymptote

Q9

$$(a) 4 \cos^2 \theta - 3 = 0, 0 \leq \theta \leq 2\pi$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

$$\therefore \theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6} \quad \checkmark \quad \left[\begin{array}{l} 3 \text{ for all} \\ 2 \text{ for } \frac{\pi}{6}, \frac{11\pi}{6} \end{array} \right]$$

$$(b) (i) \text{ if } x = y^2, \text{ let } x = 1$$

$$\therefore y = 1 \quad \therefore Q(1, 1) \quad \checkmark \quad (1)$$

$$(ii) \text{ Volume of Solid} = \pi \int_1^4 x^2 dy - \text{Volume of cylinder}$$

$$= \pi \int_1^4 y^4 dy - \pi \times 1^2 \times 3$$

$$= \pi \left[\frac{y^5}{5} \right]_1^4 - 3\pi$$

$$= \frac{\pi}{5} \times 1023 - 3\pi$$

$$= \frac{1008}{5} \pi \quad \checkmark \quad \left[\begin{array}{l} \text{ACCEPT} \\ 633.365 \end{array} \right]$$

$$(c) (i) \text{ If } Q = Q_0 e^{-kt}$$

$$\frac{dQ}{dt} = -k Q_0 e^{-kt}$$

$$= -kQ \quad \checkmark$$

$$(ii) 0.15 Q_0 = Q_0 e^{-kt} \quad \checkmark$$

$$(ii) \therefore \frac{Q_0}{2} = Q_0 e^{-4500k}$$

$$\therefore e^{-4500k} = \frac{1}{2}$$

$$-4500k = \ln \frac{1}{2}$$

$$k = -\frac{1}{4500} \ln \frac{1}{2}$$

$$= \frac{1}{4500} \ln 2 \quad \checkmark$$

$$e^{-kt} = 0.15$$

$$-kt = \ln 0.15$$

$$t = \frac{\ln 0.15}{-k}$$

$$= 17316.35 \text{ year}$$

$$= 12000 \text{ years} \quad \checkmark$$

Q10

$$(a) (i) A(x) = x \times y = x e^{-2x} \quad \checkmark$$

$$(ii) A'(x) = -2x e^{-2x} + e^{-2x}$$

$$= e^{-2x} (1 - 2x) \quad \checkmark$$

$$\text{let } A'(x) = 0 \therefore x = \frac{1}{2}$$

Test for MAX/MIN using first derivative

$x: 0.4$	$\frac{1}{2}$	0.6
$A(x): > 0$	0	< 0

MAX

$$\therefore \text{as } x = \frac{1}{2}$$

$$y = e^{-1}$$

$$\therefore A\left(\frac{1}{2}, \frac{1}{e}\right)$$

$$(b) (i) \text{ If rate} = 9\% \text{ pa}$$

$$\text{rate/month} = 0.75\%$$

$$\therefore \text{Amount owing after 1 month} = P(1.0075) - 2700$$

$$\checkmark \text{ after 2 months} = P(1.0075)^2 - 2700(1.0075)$$

$$\checkmark \text{ after 3 months} = P(1.0075)^3 - 2700(1.0075)^2 - 2700$$

$$= P(1.0075)^3 - 2700(1 + 1.0075 + 1.0075^2)$$

$$(ii) \therefore A_n = P(1.0075)^n - \frac{2700(1.0075^n - 1)}{1.0075 - 1}$$

$$\text{let } A_n = 0 \therefore P = \frac{360000(1.0075^n - 1)}{1.0075^n}$$

Q10

(b) (iii) let $n = 360$ ✓

$$\therefore P = \frac{360000(1.0075^{360} - 1)}{1.0075^{360}} \quad (2)$$

$$= \$335561.04 \quad \checkmark$$

$$\begin{aligned} \text{(iv) Interest paid} &= 360 \times 2700 - 335561.04 \\ &= \$636438.96 \quad \checkmark \end{aligned} \quad (1)$$